PULMONARY HYPERTENSION

Raghav Gattani, Amy M. Goodrich-Harris, and Jamie L.W. Kennedy

DEFINITIONS

Pulmonary hypertension (PH) is defined by the mean pulmonary artery pressure (mPAP) of >20 mm Hg at rest, which is an update from the previous 25 mm Hg or higher.[1–6] The update is based on studies which found a mPAP of 14 ± 3.3 mm Hg in normal individuals, the new definition of 20 mm Hg is 2 standard deviations above the normal mPAP.[2] Furthermore, subsequent studies involving large cohorts of patients have demonstrated a notable rise in morbidity and mortality with the mPAP exceeding 20 mm Hg.[3] As a result of these findings, a considerable subset of patients, previously considered within the normal range, have been reclassified as having PH. This change in the definition of PH highlights the importance of early diagnosis.

To fully characterize the hemodynamic derangements, it is important to integrate pulmonary capillary wedge pressure (PCWP) and pulmonary vascular resistance (PVR). PVR (in Wood units [WU]) is calculated as:

$$\text{PVR} = (\text{mPAP} - \text{mean PCWP})$$

Cardiac Output

Precapillary PH is defined by the mean PCWP of ≤15 mm Hg and PVR of ≥3 WU. Postcapillary PH is defined by the mean PCWP of >15 mm Hg and is further divided into combined pre- and postcapillary PH (CpcPH) with PVR of 3 WU or more, and isolated postcapillary PH (IpcPH) with PVR of <3 (Fig. 9.1). Prior studies had suggested that elevated transpulmonary gradient (TPG), calculated as:

$$\text{Transpulmonary gradient} = \text{mPAP} - \text{mean PCWP},$$

and diastolic pulmonary gradient, calculated as:

$$\text{Diastolic pulmonary gradient} = \text{Pulmonary artery diastolic pressure} - \text{mean PCWP}$$

could be used to differentiate IpcPH from CpcPH, but these were not included in the sixth World Symposium on PH criteria, primarily reflecting the closest association between PVR and clinical outcomes.[1]

CLASSIFICATION OF PULMONARY HYPERTENSION

The primary objective of clinical classification of PH is to group together clinical conditions that are linked by similar pathophysiologic mechanisms, hemodynamic characteristics, and therapeutic approaches. Group 1 includes pulmonary arterial hypertension (PAH) due to genetic causes, drugs/toxins, connective tissue disease, some congenital heart defects, and idiopathic pulmonary arterial hypertension (IPAH); it is characterized by precapillary hemodynamics. Group 2 is postcapillary PH, either isolated or combined, due to left heart disease such as left ventricular systolic or diastolic dysfunction, or left-sided valve disease. Group 3 consists of PH due to chronic lung disease, such as chronic obstructive pulmonary disease (COPD), interstitial lung disease, and sleep apnea, and is characterized by precapillary hemodynamics. Group 4 includes obstruction to pulmonary artery (PA) blood flow, primarily thromboembolic disease, and also typically has a precapillary hemodynamic picture. Group 5 covers multifactorial diseases such as sarcoidosis, hemolytic anemia, and complex congenital heart defects; hemodynamic profiles include precapillary PH and CpcPH[1] (Box 9.1).

DIAGNOSIS

The diagnostic strategy for patients with known or presumed PH primarily revolves around the following key objectives: (1) early clinical suspicion of PH based on history and physical exam findings, (2) initiation of suitable screening (echocardiogram) and definitive diagnostic (right heart catheterization [RHC]) methods; and (3) identification of underlying causes. This comprehensive evaluation is necessary to ensure accurate classification, risk assessment, and appropriate treatment for patients with PH.

Dyspnea is the cardinal symptom of PH and is typically experienced during physical activity in early stages of the disease. Other associated symptoms such as fatigue, bendopnea (dyspnea when bending forward), and exertional lightheadedness or syncope can be indicative of disease progression.[4] In advanced stages of PH, rare symptoms manifest. These include angina, due to either right ventricular (RV) ischemia or compression of a coronary artery by the dilated PA, and hoarseness of the voice due to compression of the left recurrent laryngeal nerve by the dilated PA. During cardiac examination, signs suggestive of PH may include a parasternal heave, an accentuated pulmonary component of the second heart sound, an S3 gallop, and a holosystolic murmur of tricuspid regurgitation (TR). Furthermore, distended jugular veins, ascites, hepatomegaly, and peripheral edema suggest right heart failure, whereas peripheral cyanosis, cool extremities, and prolonged capillary refill suggest inadequate cardiac output (CO).

PH is diagnosed by elevated pulmonary pressures on RHC; however, initial screening of patients with suspected PH is typically done with transthoracic echocardiography.[4] Doppler echocardiography can estimate pulmonary artery systolic pressure (PASP) using the maximum TR velocity and estimated right atrial pressure (RAP), as determined by the inferior vena cava diameter and collapsibility, in the following equation:

$$\text{PASP} = 4(\text{TRV}_{\text{max}}(\text{m/s}))^2 + \text{RA pressure}\left(\text{mm Hg}\right)$$

An estimated PASP greater than 40 mm Hg is generally considered abnormal, and in a patient with otherwise unexplained dyspnea, it warrants further evaluation.[5] In addition to evaluating pulmonary pressure, it is equally important to evaluate right heart size and function. Pulmonary pressure measurements can be underestimated in the setting of poor Doppler alignment or minimal tricuspid regurgitant jet, and an enlarged, hypertrophied, or dysfunctional right heart may be the only indicator of PH. Echocardiography is also useful in evaluating causes of PH, such as left heart disease and shunts. Overt left ventricular systolic dysfunction and moderate or worse left-sided valve disease suggest postcapillary PH due to left heart disease. Other signs favoring postcapillary PH on echocardiography include left ventricular hypertrophy, left atrial enlargement, grade 2 or more diastolic dysfunction, and bowing of intraatrial septum to the right. Echocardiographic findings suggestive of PH are summarized in Table 9.1.

If echocardiographic findings are suggestive of PH, a personal and family history and risk factor assessment should be investigated. A diagnostic algorithm for the diagnosis of PH is shown in Fig. 9.2. It is important to note that while echocardiography can provide valuable information, it is not the definitive diagnostic tool for PH. RHC is required to confirm the diagnosis.

RIGHT HEART CATHETERIZATION

Invasive hemodynamic evaluation with an RHC is critical for diagnosing and phenotyping PH. Performing RHC in patients with PH can be a difficult procedure and demands specialized skills, meticulous data collection, and careful attention to detail.[6] For accurate interpretation, it is necessary to have a systematic approach and follow standardized protocol in the cardiac catheterization laboratory. Additionally, comprehensive knowledge of normal and abnormal cardiac filling pressures (Table 9.2) and calculated parameters as well as recognition of pathologic waveforms, as discussed in Chapter 2, is crucial.

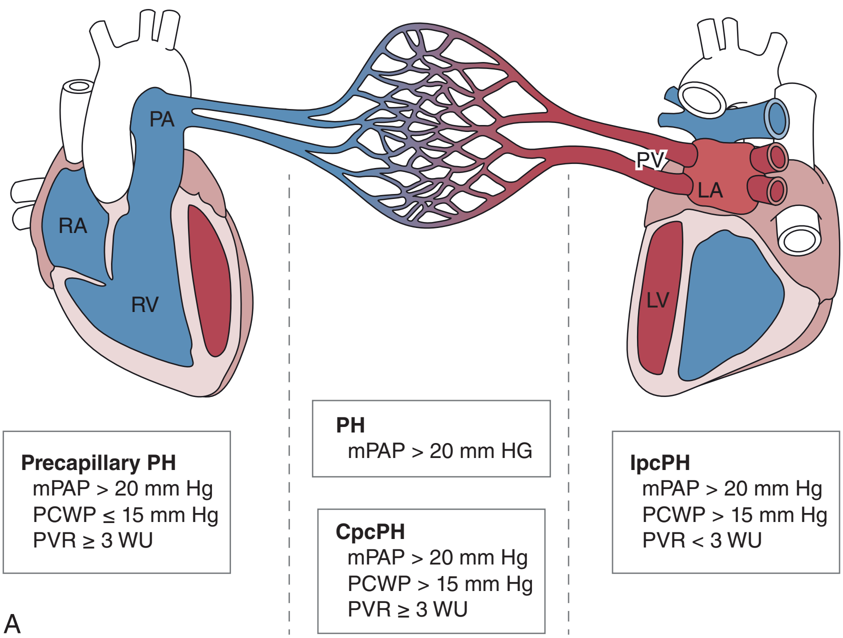
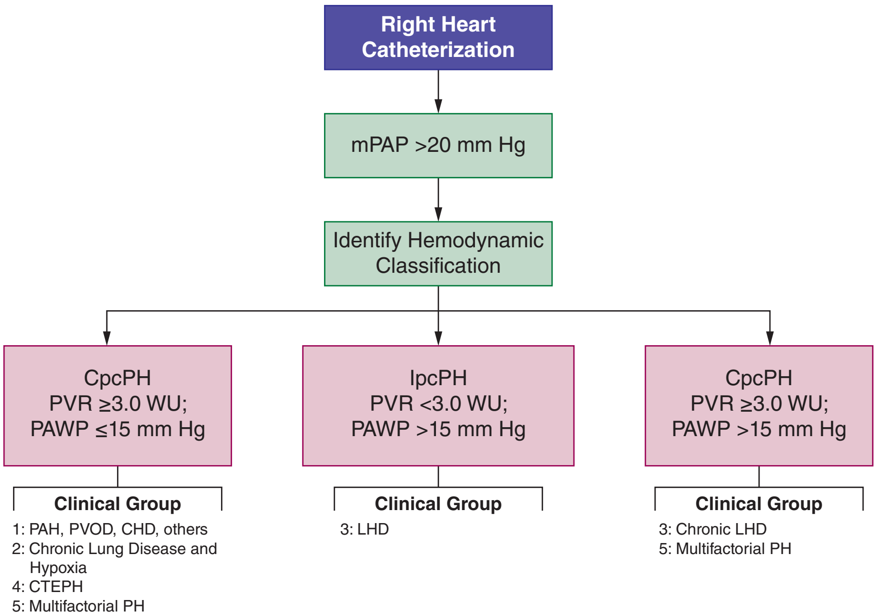


Fig. 9.1 (A) Pulmonary hypertension (PH) is classified as "precapillary" when the left atrial pressure (or pulmonary artery wedge pressure [PAWP]) is low (<15 mm Hg) and "postcapillary" when the PAWP is high (>15 mm Hg).



Hemodynamic definitions of pulmonary hypertension[a]

Definition	Hemodynamic Characteristics
PH	mPAP > 20 mm Hg
CpcPH	mPAP > 20 mm Hg PAWP ≤ 15 mm Hg PVR ≥ 3 WU
IpcPH	mPAP > 20 mm Hg PAWP > 15 mm Hg PVR < 3 WU
CpcPH	mPAP > 20 mm Hg PAWP > 15 mm Hg PVR ≥ 3 WU
Exercise PH	mPAP/CO slope between rest and exercise > 3 mm Hg/L/min

B

Fig. 9.1, cont'd (B) The classication of the subsets of PH. *CHD*, Congenital heart disease; *CO*, cardiac output; *CTEPH*, chronic thromboembolic pulmonary hypertension; *DPG*, diastolic pressure gradient; *LA*, left artery; *LHD*, left heart disease; *LV*, left vein; *mPAP*, mean pulmonary artery pressure; *PA*, pulmonary artery; *PAH*, pulmonary arterial hypertension; *PCWP*, pulmonary capillary wedge pressure; *PV*, pulmonary vein; *PVOD*, pulmonary venous occlusive disease; *PVR*, pulmonary vascular resistance; *RA*, right artery; *RV*, right ventricle; *TPG*, transpulmonary pressure gradient; *WU*, Wood units. (B, Data from Maron BA, Brittain EL, Choudhary G, Gladwin MT. Redefining pulmonary hypertension. Lancet Respir Med. 2018;6:168–170. https://www.doi.org.10.1016/S2213-2600(17)30498-8. ESC PH guidelines 2015 were updated from the data taken from ESC/ERS 2022 guidelines: https://doi.org/10.1093/eurheartj/ehac237)

After confirming an elevated mPAP (>20 mm Hg), there are two clinical questions that should be addressed by a diagnostic RHC. First is PH precapillary, isolated postcapillary, or combined pre- and postcapillary (see Fig. 9.1), and second is PH causing RV failure?

To ensure accurate pressure measurements, it is crucial to standardize its assessment. Pressure measurements should be recorded during spontaneous breathing without breath hold maneuvers due to concern for inadvertent Valsalva and preload alteration.[7] In most cases, pressure should be recorded at the end of expiration, when intrathoracic pressure is close to 0 and has minimal impact on intracardiac pressures. In patients with severe lung disease or

Box 9.1 Classification of Pulmonary Hypertension

1. Pulmonary arterial hypertension
    1. Idiopathic pulmonary arterial hypertension
    2. Heritable pulmonary arterial hypertension
        1. BMPR2
        2. *AKL-1*, *ENG*, *SMAD9, CAV1, KCNK3*
        3. Unknown
    3. Drug and toxin induced
    4. Associated with
        1. Connective tissue disease
        2. HIV infection
        3. Portal hypertension
        4. Congenital heart diseases
        5. Schistosomiasis
2. Pulmonary venoocclusive disease and/or pulmonary capillary hemangiomatosis
3. Persistent pulmonary hypertension of the newborn
4. Pulmonary hypertension due to left heart disease
    1. Left ventricular systolic dysfunction
    2. Left ventricular diastolic dysfunction
    3. Valvular disease
    4. Congenital/acquired left heart inflow/outflow tract obstruction and congenital cardiomyopathies
5. Pulmonary hypertension due to lung diseases and/or hypoxia
    1. Chronic obstructive pulmonary disease
    2. Interstitial lung disease
    3. Other pulmonary diseases with mixed restrictive and obstructive pattern
    4. Sleep-disordered breathing
    5. Alveolar hypoventilation disorders
    6. Chronic exposure to high altitude
    7. Developmental lung diseases
6. Chronic thromboembolic pulmonary hypertension
7. Pulmonary hypertension with unclear multifactorial mechanisms
    1. Hematologic disorders: Chronic hemolytic anemia, myeloproliferative disorders, splenectomy
    2. Systemic disorders: Sarcoidosis, pulmonary histiocytosis, lymphangioleiomyomatosis
    3. Metabolic disorders: Glycogen storage disease, Gaucher disease, thyroid disorders
    4. Others: Tumoral obstruction, fibrosing mediastinitis, chronic renal failure, segmental pulmonary hypertension

AKL-1, Activin receptor–like kinase 1 *BMPR2*, bone morphologic protein receptor type II; *CAV1*, caveolin-1; *ENG*, endoglin; *HIV*, human immunodeficiency virus; *KCNK3*, potassium channel subfamily K member; *SMAD9*, mothers against decapentaplegic 9.

Data from Lau E, Humber M. A critical appraisal of the updated 2014 Nice pulmonary hypertension classification system. *Can J Cardiol.* 2015;31(4):367–374; Simonneau G, Gatzoulis MA, Adatia I, et al. Updated clinical classification of pulmonary hypertension. *J Am Coll Cardiol.* 2013;62(suppl 25):D35–D41.

Table 9.1 Echocardiographic Features of Pulmonary Hypertension

Right ventricular hypertrophy
Right ventricular enlargement
RV dysfunction (reduced TAPSE and FAC)
Right atrial enlargement
D-shaped LV on short axis
Significant TR (moderate or more)
Pulmonary regurgitation (moderate or more)
Reduced RV outflow tract velocity, short acceleration time pulmonary regurgitation (moderate or more)
Dilated inferior vena cava without respiratory variation
Pericardial effusion
Dilated pulmonary arteries

LV, Left ventricle; *RV*, right ventricle; *TR*, tricuspid regurgitation; *TAPSE*, tricuspid annular plane systolic excursion; *FAC*, fractional area change.

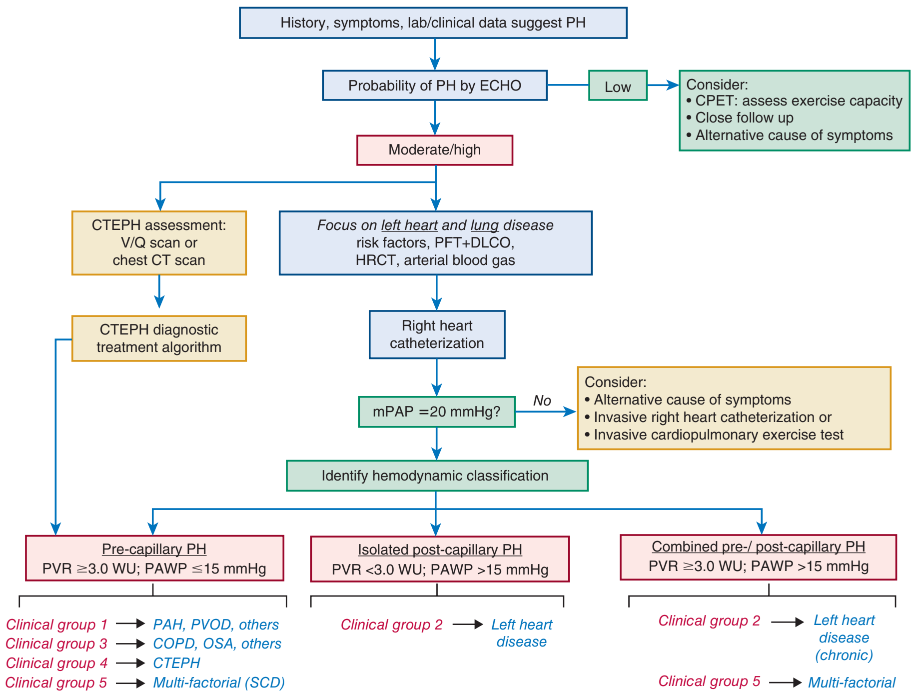


Fig. 9.2 Diagnostic algorithm for patients with dyspnea and suspected pulmonary hypertension *COPD*, Chronic obstructive pulmonary disease; *CPET*, cardiopulmonary exercise testing; *CT*, computed tomography; *CTEPH*, chronic thromboembolic pulmonary hypertension; *DLCO*, diffusion capacity of lung for carbon monoxide; *ECHO*, echocardiogram; *HRCT*, high resolution chest *CT*; *mPAP*, mean pulmonary artery pressure; *OSA*, obstructive sleep apnea; *PAH*, pulmonary arterial hypertension; *PAWP*, pulmonary artery wedge pressure; *PFT*, pulmonary function test; *PH*, Pulmonary hypertension; *PVOD*, pulmonary veno-occlusive disease; *PVR*, pulmonary vascular resistance; *SCD*, sickle cell disease; *WU*, Wood units. (From Libby P, Bonow RO, et al. *Braunwald's Heart Disease: a Textbook of Cardiovascular Medicine*. 12th ed. Elsevier Inc.; © 2022.)

morbid obesity, there may be significant respiratory pressure variation (Fig. 9.3). In such cases the end expiratory and mean values should be reported.

The PCWP is an important measurement in the hemodynamic assessment of PH as it differentiates postcapillary PH (PCWP >15 mm Hg) from precapillary PH (PCWP ≤15 mm Hg). The normal PCWP is 8.0 + 2.9 mm Hg, and abnormal is >15 mm Hg based on 2 standard deviations above the normal mean.[8] In the absence of mitral stenosis, the PCWP measured at the end of diastole, that is, the mean of the *a* wave, most closely tracks with the left ventricular end-diastolic pressure (LVEDP). The presence of large *v* waves should be noted; this strongly suggests a postcapillary component.

There are several potential pitfalls in the assessment of hemodynamics in PH. The first pitfall is when the wedge pressure tracing is actually a hybrid between the PA and PCWP tracing rather than a true PCWP tracing. During the PCWP measurement, the goal is to completely occlude the segment of PA with balloon inflation so that only downstream pulmonary venous and left atrial pressure is reflected as the PCWP. However, in severe PH, especially in PAH, dilated pulmonary arteries may preclude complete balloon occlusion. An inadequately wedged balloon enables the transmission of higher pressures from the proximal PA into the wedge tracing, thus falsely elevating it. The waveforms obtained are mixed or hybrid PA-wedge pressure waveform (Fig. 9.4) and not reflective of true PCWP. Frequently, even in fully wedged pulmonary arterial segment, as observed on fluoroscopy, severely elevated pulmonary arterial pressure may allow leakage of deoxygenated blood around the wedged balloon into the stagnant column located downstream and cause falsely elevated readings. This problem may be addressed as follows. First, measure the PCWP oxygen saturation: This is the most reliable way to prove that the measured PCWP is truly reflective of PCWP and not a hybrid PA-wedge pressure. Wedge pressure saturation should normally be ≥90% or within 5% of systemic oxygen saturation. One should always confirm the PCWP by drawing PCWP oxygen saturation. This becomes particularly crucial when evaluating PH and when the PCWP is noted to be elevated. Second, review the waveform for the timing of the peak of the V wave and the peak PASP. Close attention to the waveform can be used to differentiate the PCWP from hybrid tracings. When the waveform is correlated with the electrocardiogram, the PCWP *v* wave occurs after the T wave, while the PASP peak is during the T wave. The PCWP waveform has a horizontal or upsloping segment between the two peaks, while the PASP has a downsloping segment. Finally, consider directly measuring the LVEDP. Patients with high suspicion of diastolic heat failure and CpcPH can sometimes have borderline normal PCWP when adequately diuresed. When an

Table 9.2 Hemodynamic Measurements Obtained and Calculated During Right Heart Catheterization

	MEASURED VARIABLES	NORMAL VALUE
1.	Right atrial pressure, mean (RAP)	2–6 mm Hg
2.	Pulmonary artery pressure, systolic (sPAP)	15–30 mm Hg
3.	Pulmonary artery pressure, diastolic (dPAP)	4–12 mm Hg
4.	Pulmonary artery pressure, mean (mPAP)	8–20 mm Hg
5.	Pulmonary capillary wedge pressure, mean (PCWP)	≤15 mm Hg
6.	Cardiac output (CO)	4–8 L/min
7.	Mixed venous oxygen saturation (SvO_2)	65%–60%
CALCULATED PARAMETER		
1.	Pulmonary vascular resistance (PVR)	<3 WU
2.	RAP/PCWP ratio	<0.6
3.	Pulmonary artery pulsatility index (PAPi)	>1
4.	Pulmonary artery compliance	>2.3 mL/mm Hg

PARAMETER	EQUATION
Pulmonary vascular resistance	$\frac{\text{mPAP} - \text{PCWP}}{\text{CO}}$
RAP/PCWP ratio	$\frac{\text{RAP}}{\text{PCWP}}$
Pulmonary artery pulsatility index (PAPi)	$\frac{\text{sPAP} - \text{PCWP}}{\text{CO}}$
Pulmonary artery compliance	$\frac{\text{Stroke volume}}{\text{sPAP} - \text{dPAP}}$

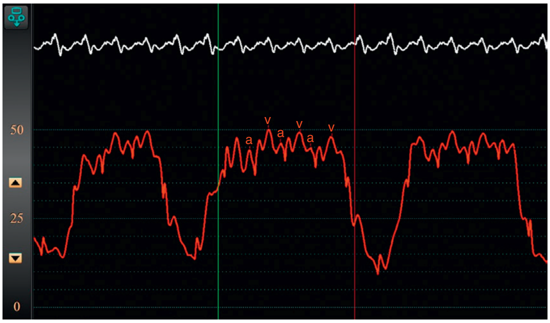


Fig. 9.3 Pulmonary capillary wedge tracing in a patient with severe left ventricular dysfunction and significantly elevated wedge pressure. This tracing shows significant respiratory variation, and the pressure should be measured at end expiration. Thus in this case, the wedge pressure is about 40 mm Hg and not 15 mm Hg. Pausing respirations briefly after exhalation during the measurement can correct this artifact.

RHC is performed in this setting, improvement of PA pressure lags behind PCWP, and hemodynamics may be falsely interpreted as isolated precapillary PH. A left heart catheterization and LVEDP measurement should be considered in this setting as the LVEDP is often higher than the PCWP in compensated LV dysfunction.

The second pitfall in hemodynamic assessment may occur in patients with PAH and abnormal, remodeled, and constricted pulmonary vasculature.[9] In these patients, the retrogradely transmitted left arterial (LA) pressure can be damped due to arteriolar and venular resistance. Although these tracings may not have distinct *a* and *v* waves, they still correlate with true PCWP.

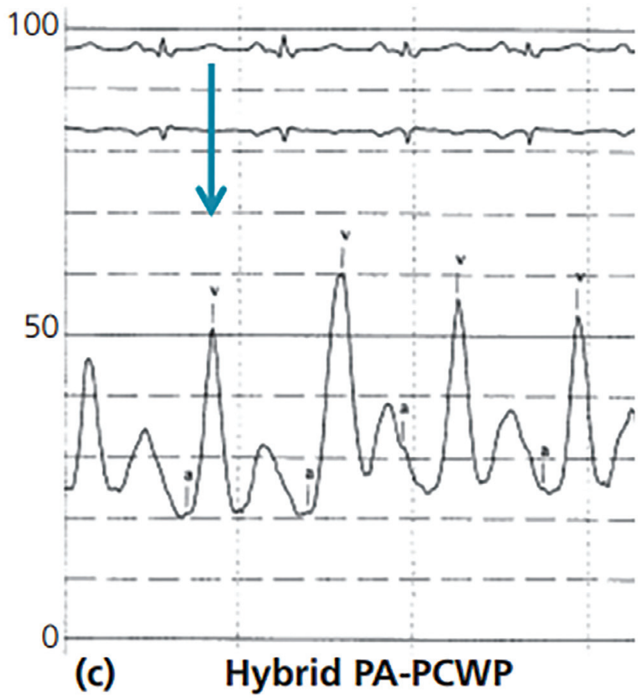


Fig. 9.4 Tracing recorded after inflation of the pulmonary artery (PA) catheter balloon. It may mimic a pulmonary capillary wedge pressure (PCWP) tracing with a large V wave. The presumed V waves peak as early as the PA pressure peaks, during the electrocardiogram T waves (*vertical arrows*), and has a sharp upslope, like the PA pressure upslope. Thus, this is not a true PCWP tracing. The PA catheter is "underwedged," which results in a hybrid PA-PCWP waveform. (From Hanna EB. Chapter 36: Hemodynamics. *Practical Cardiovascular Medicine*. © 2022 John Wiley & Sons Ltd.)

RAP Measurement

The RAP may be elevated, and the waveform may show a prominent *a* wave when contracting against higher right ventricular diastolic pressures. In decompensated right heart failure, the RAP is significantly elevated. The RA waveform may also demonstrate a prominent *v* wave representative of tricuspid valve regurgitation, which is commonly associated with PH and right heart failure. Severe regurgitation can result in obliteration of the *c* wave and replacement by a broad *v* wave termed the *c-v* wave (see Chapter 8 and also Fig. 9.5).[10] Large *v* waves can also be seen as a result of decreased atrial compliance from chronic pressure elevation in right ventricular failure. Atrial arrhythmias can often be present in patients with postcapillary PH, and less commonly in PAH. In atrial fibrillation and atrial flutter, where there is no organized atrial activity, the right atrial waveform will show a loss of *a* waves.

Right Ventricular Pressure Measurement

In PH with normal right ventricular function, the right ventricular systolic pressure is elevated to pump against increased pulmonary pressures. Under these conditions, the right ventricular end-diastolic pressure (RVEDP) will remain normal. As the RV fails, the RVEDP rises. The right ventricular pressure drops rapidly in early diastole during isovolumic relaxation, and on the right ventricular pressure waveform, there will be a sharp early diastolic dip with a rise and plateau of elevated diastolic pressure. Additionally, a prominent *a* wave may be seen, demonstrating noncompliance of the RV (Fig. 9.6).

PA Pressure Measurement

The PASP will be elevated and should be equal to the right ventricular systolic pressure in the absence of pulmonic valve stenosis. The PA diastolic pressure is an indirect measurement of the LA and LVEDP in the absence of downstream pulmonary vasculature or mitral valve obstruction. In postcapillary PH (group 2), the pulmonary diastolic pressure will also be elevated (Fig. 9.7). In very severe disease, the pulmonary pressures may decrease to only modestly elevated levels because of the inability of the failing RV to generate high pressures. This will be seen in combination with a markedly elevated RAP and RVEDP, and low CO. PVR remains significantly elevated as well.

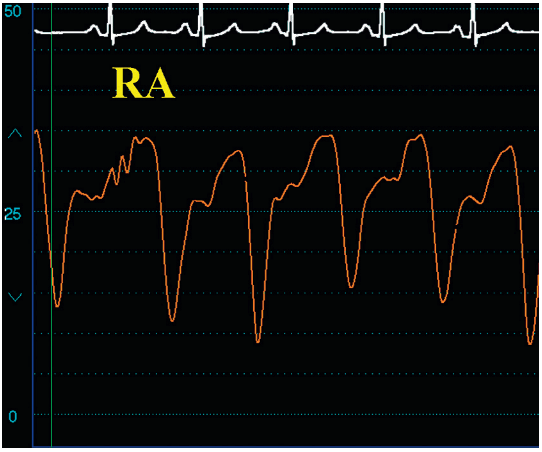


Fig. 9.5 Right atrial (RA) pressure waveform obtained in a patient with severe tricuspid regurgitation showing the absence of the “x” descent and a prominent “c-v” wave and prominent “y” descent.

Fig. 9.6 Right ventricular pressure waveform demonstrating an early diastolic dip and an *a* wave (see *circled a* wave in figure) representative of ventricular noncompliance, as well as elevated systolic and end-diastolic pressures. *d*, Diastole; *e*, end diastole; *s*, systole.

Fig. 9.7 Pulmonary artery pressure tracing showing severely elevated pulmonary pressures reaching systemic levels and also elevation of the pulmonary artery diastolic pressures.

CO Measurement

While the direct Fick method is regarded as the most accurate way to estimate CO, it necessitates specialized equipment to accurately measure the oxygen consumption and is not widely accessible in most laboratories. Therefore CO is most frequently assessed using either the indirect Fick method, which is often unreliable in patients with heart failure, undiagnosed shunts, PH, and obesity, or the more dependable thermodilution (TD) method. Despite being historically regarded as less accurate in severe TR and low CO conditions, TD outperforms the indirect Fick method and has shown good correlation with the direct Fick method.[7,10] Additionally, it is worth noting that patients with high CO state such as morbid obesity, anemia, and hypoxemia can exacerbate PH. [11]

Pulmonary Vascular Resistance

From the hemodynamic measurements, further calculations can be made to determine the PVR. PVR in WU is calculated using Ohm's law as follows:

$$\text{PVR} = \frac{\text{mPAP (mm Hg)} - \text{PCWP (mm Hg)}}{\text{Cardiac output (L/min)}}$$

Traditionally, a PVR of >3 WU is associated with isolated precapillary and CpcPH; recent data suggest a threshold of 2.2 WU to be clinically meaningful in predicting worse outcomes.[12] In PAH, the two hemodynamic variables most closely correlated with outcomes are RAP and PVR.[13] As mentioned, pulmonary pressures may fall late in the disease course due to right heart failure, while PVR continues to climb (Fig. 9.8).

Shunt Assessment

All patients with newly diagnosed precapillary PH should be assessed for the presence of a left to right intracardiac shunt. In routine practice, it can be rather cumbersome to perform a complete shunt run in all patients with PH. A good screening method is to compare the oxygen saturation from blood samples obtained from the superior vena cava and from the main PA. A step up of $\geq$7% is indicative of a significant left to right shunt and would warrant a more comprehensive saturation run to identify the source and size of the shunt as detailed in Chapter 3. Of note, a direct Fick method is the preferred method of CO measurement if an intracardiac shunt is suspected.

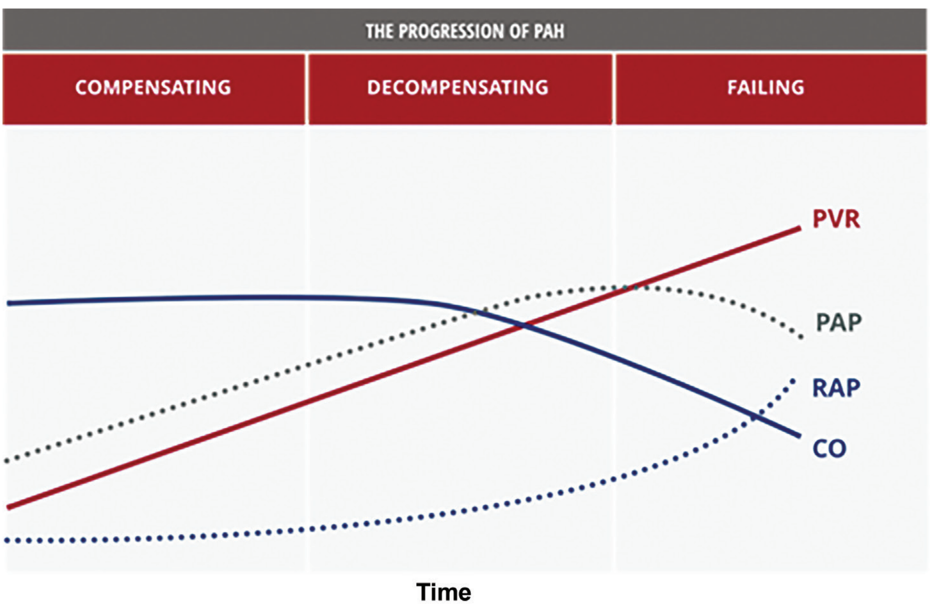


Fig. 9.8 Evolution of hemodynamic parameters with disease progression. *CO*, Cardiac output; *PAH*, pulmonary arterial hypertension; *PVR*, pulmonary vascular resistance; *RAP*, right atrial pressure; *PAP*, pulmonary artery pressure. (From Hill NS. Historical perspective and classification. In: Hill NS, ed. Pulmonary Hypertension Therapy. Summit Communications, LLC; 2006:9.)

PROVOCATIVE TESTING

A resting RHC may be inadequate in patients with exertional dyspnea with otherwise normal hemodynamics. During the initial phases of cardiopulmonary disease or as a result of diuretic use, the resting hemodynamics may appear normal despite the presence of underlying pulmonary vascular or cardiac abnormalities. In such cases, hemodynamic abnormalities may only become apparent with provocation. RHC with provocative maneuvers such as dynamic exercise or saline loading can be particularly important in the evaluation of undifferentiated dyspnea (Fig. 9.9). These maneuvers can unmask occult PH due to left heart disease (PH-LHD) and help to differentiate PAH from PH-LHD[6,7,14] (Fig. 9.10). Additionally, vasodilator challenges can be performed to assess the reactivity of pulmonary vasculature in selected patients with group 1, to guide management, or group 2 PH, as part of cardiac transplant candidacy evaluation.

Exercise RHC

Exercise RHC can be challenging to perform due to both patient related and technical factors. During exercise there are large swings in intrathoracic pressure that can overestimate end expiratory pressure measurements. Also, exercise can cause increased motion artifacts and can confound accurate assessment of waveforms. Fluid challenge provides more reproducible results outside of centers with significant experience conducting exercise RHC studies. Protocols for exercise RHC may differ across centers but should include an escalating exercise workload in supine position, and measurements of mPAP, systolic pulmonary artery pressure, diastolic PAP, PCWP, CO, systemic oxygen saturation, and systemic blood pressure at each exercise level. Of note, exercise increases the peripheral oxygen consumption, resulting in a fall in pulmonary arterial oxygen saturation and rendering calculation of CO by the indirect Fick technique useless, TD or direct Fick techniques should be utilized instead. During catheterization, exercise is best done with a supine bicycle, so that the patient does not need to move from the catheterization table. Exercise should start at a 20 Watts workload for 5 minutes and increase to 10 Watts increments every 3 minutes until the patient reports exhaustion.

Prior work suggested exercise-induced PH may be an early form of precapillary PH. More recently it has been recognized that postcapillary PH also presents with exercise-induced PH and is by far more common. The relationship between CO and mPAP is similar between patients with precapillary and postcapillary PH and therefore cannot be used to reliably differentiate the two.[13] An exercise PCWP of >25 mm Hg was previously used to define occult PH due to left heart disease. Concomitant increase in PVR to >2 WU suggests mixed pre- and postcapillary PH. Recently, a new parameter, ΔPCWP/ΔCO slope >2 mm Hg/L/min, has been used to more reliably suggest exercise-induced PH-LHD/HfpEF (Fig. 9.11).[14] In healthy subjects, the PCWP increases linearly with CO during exercise, thus maintaining a ratio <2. On the contrary, in patients with occult HfpEF, the PCWP rises dramatically during exercise but with relatively muted increase in CO, thus raising the slope of PCWP/CO to >2 with exercise. Exertional hypoxia correlates strongly with precapillary PH and is quite uncommon in patients with postcapillary PH; this test does not necessarily require a trip to the cath lab.[15]

Saline loading

Compared with exercise RHC, rapid saline loading during RHC is less complex technically and requires no specialized equipment. An abnormal test is indicated by a PCWP of >18 mm Hg after 500 cc of normal saline infusion over 5 minutes and is suggestive of PH-LHD.[16]

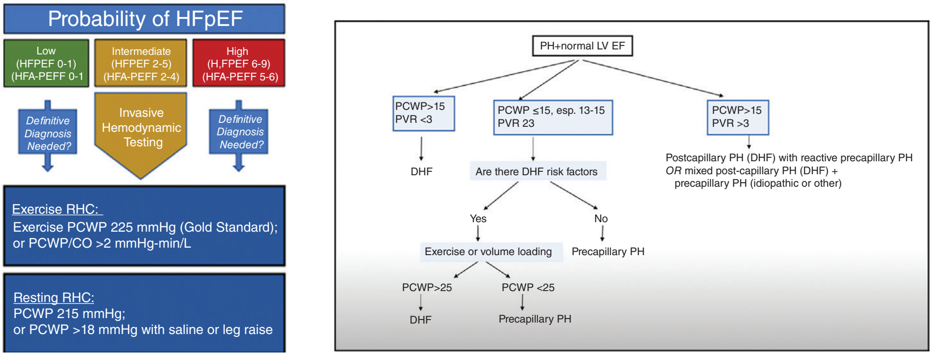


Fig. 9.9 Role of provocative maneuvers in the evaluation of pulmonary hypertension (PH). *CO*, Cardiac output; *DHF*, diastolic heart failure; *HFpEF*, heart failure with preserved ejection fraction; *LVEF*, left ventricular ejection fraction; *PCWP*, pulmonary capillary wedge pressure; *PVR*, pulmonary vascular resistance; *RHC*, right heart catheterization. (Left, From Hsu S, Fang JC, Borlaug BA. Hemodynamics for the heart failure clinician: a state-of-the-art review. *J Card Fail*. 2022;28(1):133–148. https://www.doi.org.10.1016/j.cardfail.2021.07.012. Epub 2021 Aug 10. PMID: 34389460; PMCID: PMC8748277. Right, From YouTube channel by Elias Hanna [Hemodynamics].)

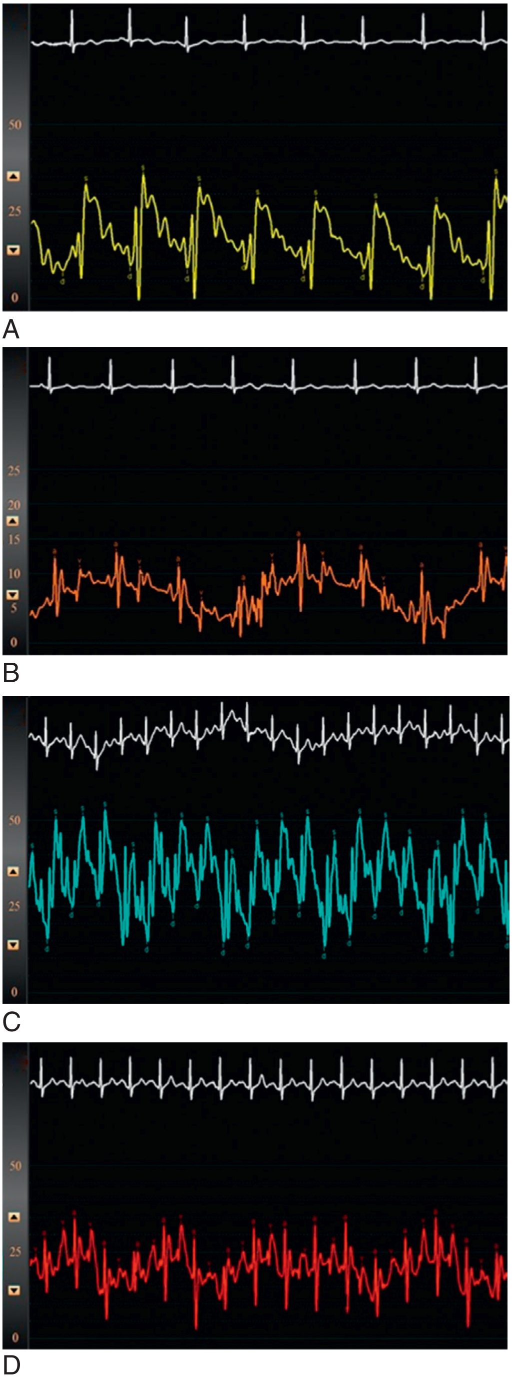


Fig. 9.10 This patient had (A) normal pulmonary artery and (B) normal pulmonary capillary wedge pressures at rest. After exercise, both (C) pulmonary artery and (D) pulmonary capillary wedge pressures are elevated. The patient had normal left ventricular systolic function and symptoms of heart failure. These tracings suggest the diagnosis of heart failure with preserved ejection fraction. *a*, Atrial contraction; *d*, diastole; *s*, systole; *v*, venous filling.

1. Baseline hemodynamic measurements:
   a. RA pressure.
   b. RV pressure.
   c. PA pressure.
   d. PCWP.
   e. Cardiac output and index (thermodilution and Fick methods).
2. If the etiology of PH is unknown or PA saturation is >75%, perform a shunt run.
3. If the PCWP is in question, measure left-ventricular end-diastolic pressure, or confirm PCWP with saturation (≥94%).
4. If the baseline hemodynamics are normal or the symptoms are out of proportion to the hemodynamics, then consider exercise testing or volume challenge to identify diastolic dysfunction.
5. If the PASP is >50 mmHg, mPAP >25 mmHg, TPG >15 mmHg, or PVR >3 Wood units, then perform a vasodilator challenge.
   a. If the PCWP is <18 mmHg, use iNO at 20 ppm for 5 minutes.
   b. If the PCWP is ≥18 mmHg, use sodium nitroprusside infusion.
      i. Start at 0.25 mcg/kg per min.
      ii. Increase by 0.25 mcg/kg per min every 2 minutes until one of the following occurs:
         1. Maximum dose of 10 mcg/kg per min is reached.
         2. Systemic mean arterial pressure decreases to <60 mmHg.
         3. PVR drops to <2 Wood units.
6. Repeat the PA pressure, PCWP, and cardiac output and index (thermodilution and Fick methods) while still on vasodilating agents (iNO or sodium nitroprusside).
7. After repeat measurements are made, stop iNO/sodium nitroprusside.

Fig. 9.11 Example of a hemodynamic and vasodilator challenge protocol for evaluation of patients with pulmonary hypertension (PH). *iNO*, Inhaled nitric oxide; *mPAP*, mean pulmonary artery pressure; *PA*, pulmonary artery; *PASP*, pulmonary artery systolic pressure; *PCWP*, pulmonary capillary wedge pressure; *PVR*, pulmonary vascular resistance; *RA*, right atrial; *RV*, right ventricle; *TPG*, transpulmonary gradient.

PAH vasodilator testing

Pulmonary vasodilator testing to identify patients suitable for high-dose calcium channel blocker (CCB) treatment is recommended for patients with IPAH, hereditary PAH, and drug-induced PAH. In other forms of PAH or PH, the results could be misleading, and often these patients are not responders. Improvement with acute vasodilator testing, defined as a decrease in mPAP by at least 10 mm Hg to an absolute level of less than 40 mm Hg without a decrease in CO, correlates with better long-term prognosis and response to treatment with CCBs.[17] Of note, patients with poor hemodynamics such as systemic hypotension, low CO, and/or elevated RAP represent a high-risk population and should not be treated with CCB regardless of their response to acute vasodilator challenge. Inhaled nitric oxide is the preferred and most used vasodilator agent. It is the most specific and short-acting vasodilator for pulmonary vessels and has no systemic effects. Other available agents include intravenous epoprostenol and intravenous adenosine. CCB, sodium nitroprusside, and nitrates should not be used in PAH for vasodilator testing (Table 9.3).

Pulmonary venoocclusive disease (PVOD) is a rare form of PH caused by obliteration of small pulmonary veins by fibrous intimal thickening and patchy capillary proliferation. Hemodynamically, it is indistinguishable from PAH. With vasodilators, there is preferential dilation of pulmonary arterioles and flooding of pulmonary capillaries causing pulmonary edema. This may manifest acutely in the cath lab during vasodilator challenge, or subacutely after several weeks of treatment with pulmonary vasodilators.[18] PVOD carries a poor prognosis, and potentially appropriate candidates should be referred for lung transplant evaluation early in their course.

CPCPH VASODILATOR TESTING

Evaluation of the pulmonary vasculature is critical in patients being evaluated for heart transplantation. Right heart failure accounts for approximately 20% of early posttransplant deaths. Risk of right heart failure following heart transplant is considered prohibitively high if PVR is >5 WU, PVRI >6 (TPG/CI), or TPG >16 mm Hg, particularly if PASP is >60 mm Hg.

Table 9.3 Agents for Vasodilator Testing

	NITRIC OXIDE	ADENOSINE	EPOPROSTENOL
Route of administration	Inhaled	Intravenous infusion	Intravenous infusion
Dose titration	None	50 mcg/kg per min every 2 min	2 ng/kg per min every 10–15 min
Dose range	10–80 ppm	50–250 mcg/kg per min	2–10 ng/kg per min
Side effects	Increased left heart filling pressure in susceptible patients (PCWP >18 mm Hg)	Dyspnea, chest pain, AV block	Headache, nausea, lightheadedness

From McLaughlin VV, Archer SL, Badesch DB, et al. ACCF/AHA 2009 expert consensus document on pulmonary hypertension: a report of the American College of Cardiology Foundation Task Force on Expert Consensus Documents and the American Heart Association developed in collaboration with the American College of Chest Physicians; American Thoracic Society, Inc.; and the Pulmonary Hypertension Association. *J Am Coll Cardiol.* 2009;53:1573–1619.
AV, Atrioventricular; *PCWP*, pulmonary capillary wedge pressure.

Improvement in PVR to less than 2.5 WU with a vasodilator trial reduces the risk of right heart failure, unless accompanied by systemic hypotension (systolic blood pressure < 85 mm Hg).[19] Vasodilator challenge with sodium nitroprusside is recommended if the PA systolic pressure is greater than or equal to 50 mm Hg and PVR is greater than 3 WU. An example of a protocol for vasodilator challenge in a patient with PH is shown in Fig. 9.12, and an example of a patient with PH due to left heart disease undergoing a hemodynamic and vasodilator challenge protocol using sodium nitroprusside is shown in Fig. 9.11.

TREATMENT AND EFFECTS ON HEMODYNAMICS

Treatment of PH, specifically PAH, has evolved in the past decade. The goals of treatment are to improve symptoms, enhance functional capacity, lower PA pressures, normalize CO, prevent progression of disease, and improve survival (Fig. 9.13).

For PAH patients, targeted therapies are available and include endothelin receptor antagonists (ambrisentan, bosentan, and macitentan), phosphodiesterase type 5 inhibitors (sildenafil and tadalafil), guanylate cyclase stimulators (riociguat), prostacyclin analogs (epoprostenol, iloprost, and treprostinil), and selective prostacyclin receptor agonists (selexipag).[6] Through various mechanisms, treatment with these medications results in pulmonary vasodilation. In clinical studies, these medications have all been shown to decrease PVR. Initial combination therapy is now considered standard of care for the vast majority of patients. For patients with poor functional class, overt right heart failure, or other high-risk features, initial parenteral therapy should be considered. If medical therapy is ineffective, appropriate candidates should be evaluated for lung transplantation.[20]

The role of pulmonary vasodilator therapy in patients with other forms of PH (groups 2–5) is less clearly defined. In patients with postcapillary PH, initial treatment focuses on the underlying left heart disease. In patients with CpcPH, patients should be evaluated for contributing factors, such as sleep-disordered breathing, thromboembolic disease, connective tissue diseases, HIV, and drug/toxin exposure (especially methamphetamine use). If patients fail to improve with the above measures, appropriate candidates should be considered for heart transplantation. Left ventricular systolic heart failure patients managed with left ventricular assist devices, either percutaneous or durable, can demonstrate substantial improvement in pulmonary pressures with adequate LV unloading and time, permitting eventual heart transplantation. Patients not eligible for left ventricular assist devices and significantly elevated PVR may require combined heart and bilateral lung transplant.

Treatments for group 3 PH (due to lung disease and/or hypoxia) similarly focus initially on the underlying disease. In patients with COPD, oxygen supplementation has been shown to reduce progression of PH; patients with interstitial lung disease should also be treated with supplemental oxygen if needed, although the benefit on PH is less clear. Sleep-disordered breathing should be treated to prevent episodic hypoxia and progression of PH. There has been long-standing concern regarding the use of systemic pulmonary vasodilators in patients with chronic lung disease resulting in worsening ventilation/perfusion mismatch and therefore systemic hypoxia. Inhaled pulmonary vasodilators are attractive in this regard, as they are primarily delivered to well-ventilated parts of the lung. Treatment with inhaled treprostinil in patients with interstitial lung disease resulted in improved exercise capacity and delay in clinical worsening.[21]

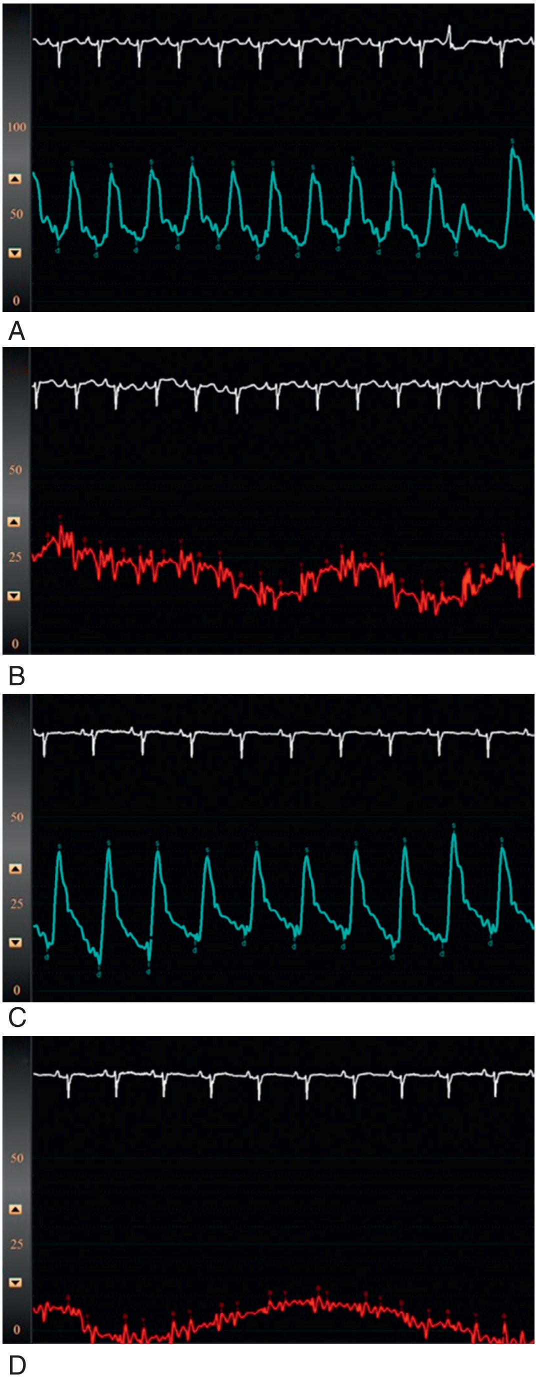
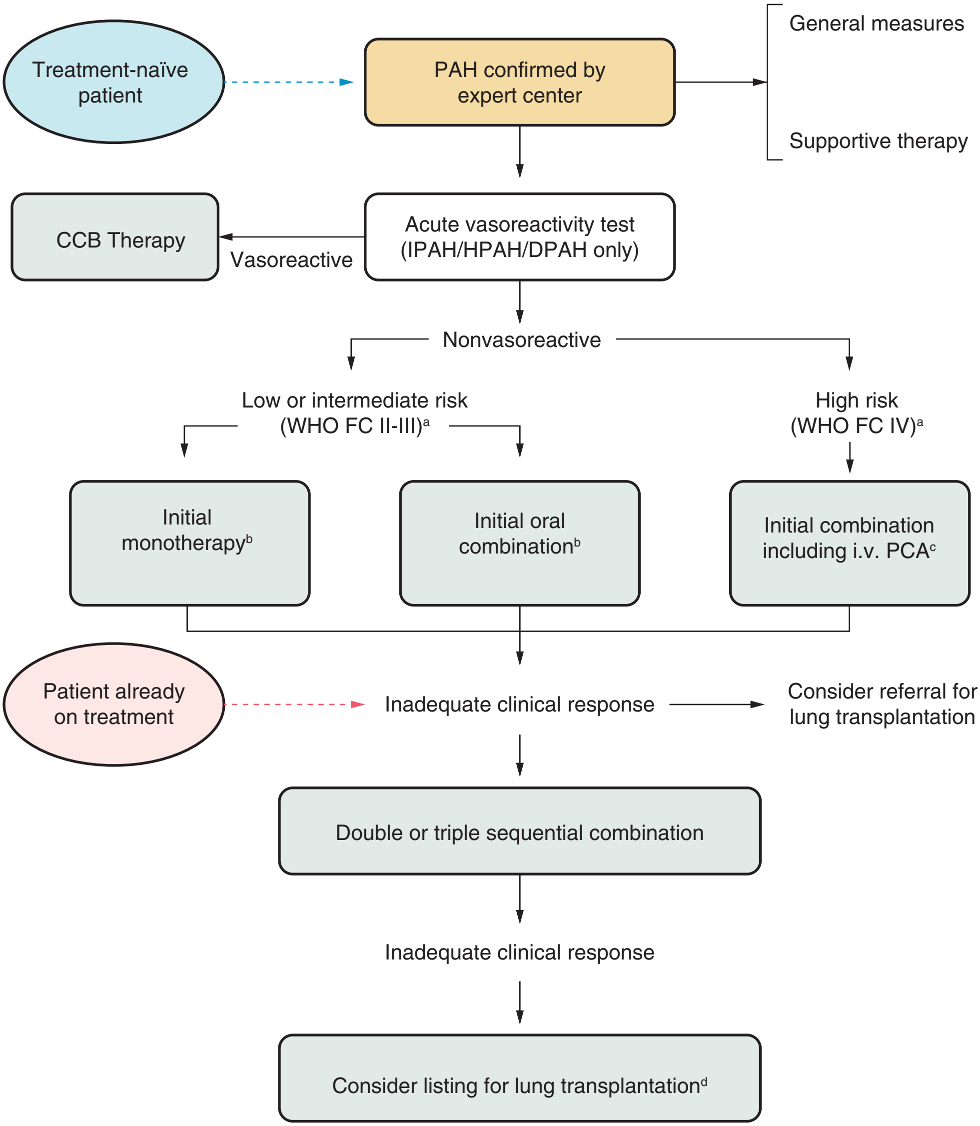


Fig. 9.12 Vasodilator challenge with sodium nitroprusside in a patient with left heart disease. (A) Baseline pulmonary artery (PA) pressure of 76/35 mm Hg. (B) Baseline pulmonary capillary wedge pressure (PCWP) of 20/25 mm Hg. (C) PA pressure decreased by 24 mm Hg postintravenous sodium nitroprusside to 40/12 mm Hg. (D) PCWP decreased by 17 mm Hg postintravenous sodium nitroprusside to 5 mm Hg. *a,* Atrial contraction; *d,* diastole; *s,* systole; *v,* venous filling.



[a]Some WHO-FC III patients may be considered high-risk.
[b]Initial combination with ambrisentan plus tadalafil has proven to be superior to initial monotherapy with ambrisentan or tadalafil in delaying clinical failure.
[c]Intravenous epoprostenol should be prioritized as it has reduced the 3 month rate for mortality in high-risk PAH patients also as monotherapy.
[d]Consider also balloon atrial septostomy.

Fig. 9.13 Pulmonary arterial hypertension (PAH) treatment algorithm. *CCBs*, Calcium channel blockers; *DPAH*, drug-induced pulmonary arterial hypertension; *ERAs*, endothelin receptor antagonists; *FC*, functional class; *HPAH*, hereditary pulmonary arterial hypertension; *IPAH*, idiopathic pulmonary arterial hypertension; *IV*, intravenous; *LOE*, level of evidence; *PDE-5i*, phosphodiesterase-5 inhibitor; *PH*, pulmonary hypertension; *SC*, subcutaneous; *WHO*, World Health Organization. (Redrawn from Galiè N, Humbert M, Vachiery JL, et al. 2015 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension: the Joint Task Force for the Diagnosis and Treatment of Pulmonary Hypertension of the European Society of Cardiology (ESC) and the European Respiratory Society (ERS): Endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC), International Society for Heart and Lung Transplantation (ISHLT). *Eur Heart J.* 2016;37:67–119.)

In group 4 chronic thromboembolic pulmonary hypertension (CTEPH), lifelong therapeutic anticoagulation is warranted. Pulmonary endarterectomy is the treatment of choice in this group and is potentially curative. Surgical success requires careful patient selection and operator experience. With successful surgery, patients often experience substantial symptom relief and dramatic improvement or even normalization of hemodynamics.[6] Patients with CTEPH should be evaluated at an experienced pulmonary endarterectomy center as soon as possible after diagnosis. Of course, not all patients will be candidates for surgery, either due to surgically inaccessible thromboembolic disease or due to comorbid conditions. In nonoperable patients, or those with persistent PH after endarterectomy, targeted medical therapy is indicated. Riociguat is the best studied pulmonary vasodilator in this patient population, with demonstrated improvement in functional class and exercise capacity.[22] In patients with inoperable CTEPH and persistent PH on medical therapy, balloon pulmonary angioplasty (BPA) has become a viable treatment modality.[23] BPA can reduce PVR by ~50% and improves right heart function and exercise capacity.

Group 5 PH includes disorders with unclear or multifactorial mechanisms, such as sickle cell anemia, chronic kidney disease, sarcoidosis, and others. Treatment for this group is tailored to the underlying disease process, and there are no randomized controlled trials evaluating the use of PAH drugs in this group as a whole. A double-blind, placebo-controlled trial comparing sildenafil to placebo in patients with sickle cell disease showed no improvement in 6-minute walk distance, and there was an increased rate of hospitalization for pain in patients treated with sildenafil.[24] Prostacyclins, endothelin receptor antagonists, and phosphodiesterase inhibitors have all been used off-label in patients with pulmonary sarcoidosis and have shown some hemodynamic and clinical improvement in this population. No large randomized controlled trials have been done to approve their use in these patients.

Patients with mild PH (mPAP 20–24 mm Hg) are associated with an increased risk of all-cause mortality. However, since this group was not included in previous clinical trials, there are many unanswered questions regarding their evaluation and management. Preliminarily, the primary objective of clinical management should be to identify the underlying cause of mild PH, modify risk factors, closely monitor symptom progression, and carefully consider the possibility of enrolling these patients in clinical trials.

References

1. Galiè N, Humbert M, Vachiery JL, et al. 2015 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension: the Joint Task Force for the Diagnosis and Treatment of Pulmonary Hypertension of the European Society of Cardiology (ESC) and the European Respiratory Society (ERS): Endorsed by: Association for European Paediatric and Congenital Cardiology (AEPC), International Society for Heart and Lung Transplantation (ISHLT). *Eur Heart J*. 2016;37:67–119.
2. Kovacs G, Berghold A, Scheidl S, et al. Pulmonary arterial pressure during rest and exercise in healthy subjects: a systematic review. *Eur Respir J*. 2009;34:888–894.
3. Maron BA, Brittain EL, Choudhary G, Gladwin MT. Redefining pulmonary hypertension. *Lancet Respir Med*. 2018;6:168–170. https://doi.org/10.1016/S2213-2600(17)30498-8.
4. Frost A, Badesch D, Gibbs JSR, et al. Diagnosis of pulmonary hypertension. *Eur Respir J*. 2019;53(1):1801904. https://doi.org/10.1183/13993003.01904-2018. PMID: 30545972; PMCID: PMC6351333.
5. Kanwar MK, Tedford RJ, Thenappan T, De Marco T, Park M, McLaughlin V. Elevated pulmonary pressure noted on echocardiogram: a simplified approach to next steps. *J Am Heart Assoc*. 2021;10(7):e017684. https://doi.org/10.1161/JAHA.120.017684. Epub 2021 Mar 15. PMID: 33719491; PMCID: PMC8174323.
6. Humbert M, Kovacs G, Hoeper MM, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Respir J*. 2023 Jan 6;61(1):2200879. https://doi.org/10.1183/13993003.00879-2022. PMID: 36028254.
7. Maron BA, Kovacs G, Vaidya A, et al. Cardiopulmonary hemodynamics in pulmonary hypertension and heart failure: JACC review topic of the week. *J Am Coll Cardiol*. 2020;76(22):2671–2681. https://doi.org/10.1016/j.jacc.2020.10.007. PMID: 33243385; PMCID: PMC7703679.
8. Vachiéry JL, Tedford RJ, Rosenkranz S, et al. Pulmonary hypertension due to left heart disease. *Eur Respir J*. 2019;53(1):1801897. https://doi.org/10.1183/13993003.01897-2018. PMID: 30545974; PMCID: PMC6351334.
9. Tuder RM. Pulmonary vascular remodeling in pulmonary hypertension. *Cell Tissue Res*. 2017;367(3):643–649. https://doi.org/10.1007/s00441-016-2539-y. Epub 2016 Dec 26. PMID: 28025704; PMCID: PMC5408737.
10. Rosenkranz S, Preston IR. Right heart catheterization: best practice and pitfalls in pulmonary hypertension. *Eur Respir Rev*. 2015;24(138):642–652. https://doi.org/10.1183/16000617.0062-2015. PMID: 26621978; PMCID: PMC9487613.
11. Friedman SE, Andrus BW. Obesity and pulmonary hypertension: a review of pathophysiologic mechanisms. *J Obes*. 2012;2012:505274. https://doi.org/10.1155/2012/505274. Epub 2012 Sep 3. PMID: 22988490; PMCID: PMC3439985.
12. Simonneau G, Montani D, Celermajer DS, et al. Haemodynamic definitions and updated clinical classification of pulmonary hypertension. *Eur Respir J*. 2019;53(1):1801913. https://doi.org/10.1183/13993003.01913-2018. PMID: 30545968; PMCID: PMC6351336.
13. Herve P, Lau EM, Sitbon O, et al. Criteria for diagnosis of exercise pulmonary hypertension. *Eur Respir J*. 2015;46(3):728–737. https://doi.org/10.1183/09031936.00021915. Epub 2015 May 28. PMID: 26022955.
14. Tea I, Hussain I. Under pressure: right heart catheterization and provocative testing for diagnosing pulmonary hypertension. *Methodist Debakey Cardiovasc J*. 2021;17(2):92–100. https://doi.org/10.14797/AFUI4711. PMID: 34326928; PMCID: PMC8298122.
15. Hardin KM, Giverts I, Campain J, et al. Systemic arterial oxygen levels differentiate pre- and post-capillary predominant hemodynamic abnormalities during exercise in undifferentiated dyspnea on exertion. J Card Fail. 2023:S1071–9164(23)00244-0. https://www.doi.org.10.1016/j.cardfail.2023.05.023. Epub ahead of print. PMID: 37467924.
16. D'Alto M, Badesch D, Bossone E, et al. A fluid challenge test for the diagnosis of occult heart failure. *Chest*. 2021;159(2):791–797. https://doi.org/10.1016/j.chest.2020.08.019.
17. Halliday SJ, Hemnes AR, Robbins IM, et al. Prognostic value of acute vasodilator response in pulmonary arterial hypertension: beyond the "classic" responders. *J Heart Lung Transplant*. 2015;34(3):312–318. https://doi.org/10.1016/j.healun.2014.10.003. Epub 2014 Nov 4. PMID: 25577565; PMCID: PMC4380573.

18. Montani D, Lau EM, Dorfmüller P, et al. Pulmonary veno-occlusive disease. *Eur Respir J*. 2016;47(5):1518–1534. https://doi.org/10.1183/13993003.00026-2016. Epub 2016 Mar 23. PMID: 27009171.
19. Velleca A, Shullo MA, Dhital K, et al. The International Society for Heart and Lung Transplantation (ISHLT) guidelines for the care of heart transplant recipients. *J Heart Lung Transplant*. 2023;42(5):e1–e141. https://doi.org/10.1016/j.healun.2022.10.015. Epub 2022 Dec 20. PMID: 37080658.
20. Galiè N, Barberà JA, Frost AE, et al. Initial use of ambrisentan plus tadalafil in pulmonary arterial hypertension. *N Engl J Med*. 2015;373(9):834–844. https://doi.org/10.1056/NEJMoa1413687. PMID: 26308684.
21. Waxman A, Restrepo-Jaramillo R, Thenappan T, et al. Inhaled treprostinil in pulmonary hypertension due to interstitial lung disease. *N Engl J Med*. 2021;384:325–334. https://doi.org/10.1056/NEJMoa2008470.
22. Ghofrani HA, D'Armini AM, Grimminger F, et al. Riociguat for the treatment of chronic thromboembolic pulmonary hypertension. *N Engl J Med*. 2013;369(4):319–329. https://doi.org/10.1056/NEJMoa1209657. PMID: 23883377.
23. Coghlan JG, Rothman AM, Hoole SP. Balloon pulmonary angioplasty: state of the art. *Interv Cardiol*. 2021;16:e02. PMID: 33664801; PMCID: PMC7903587.https://doi.org/10.15420/icr.2020.14.
24. Galiè N, Channick RN, Frantz RP, et al. Risk stratification and medical therapy of pulmonary arterial hypertension. *Eur Respir J*. 2019;53(1):1801889. https://doi.org/10.1183/13993003.01889-2018. PMID: 30545971; PMCID: PMC6351343.